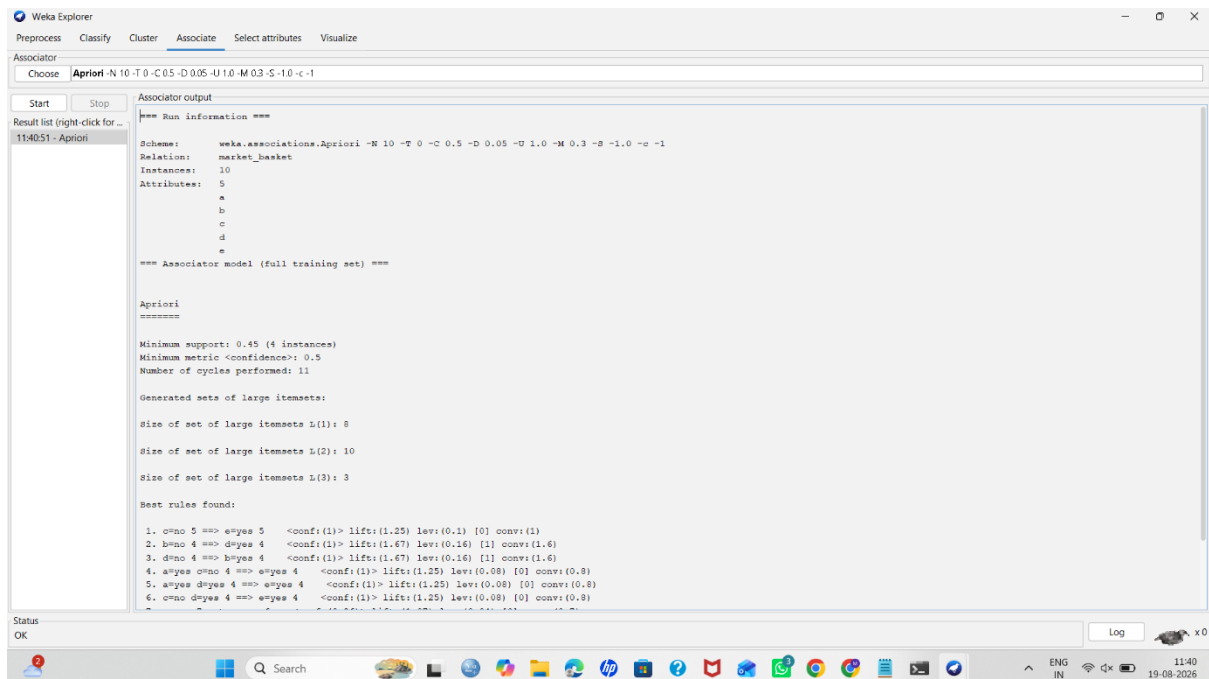
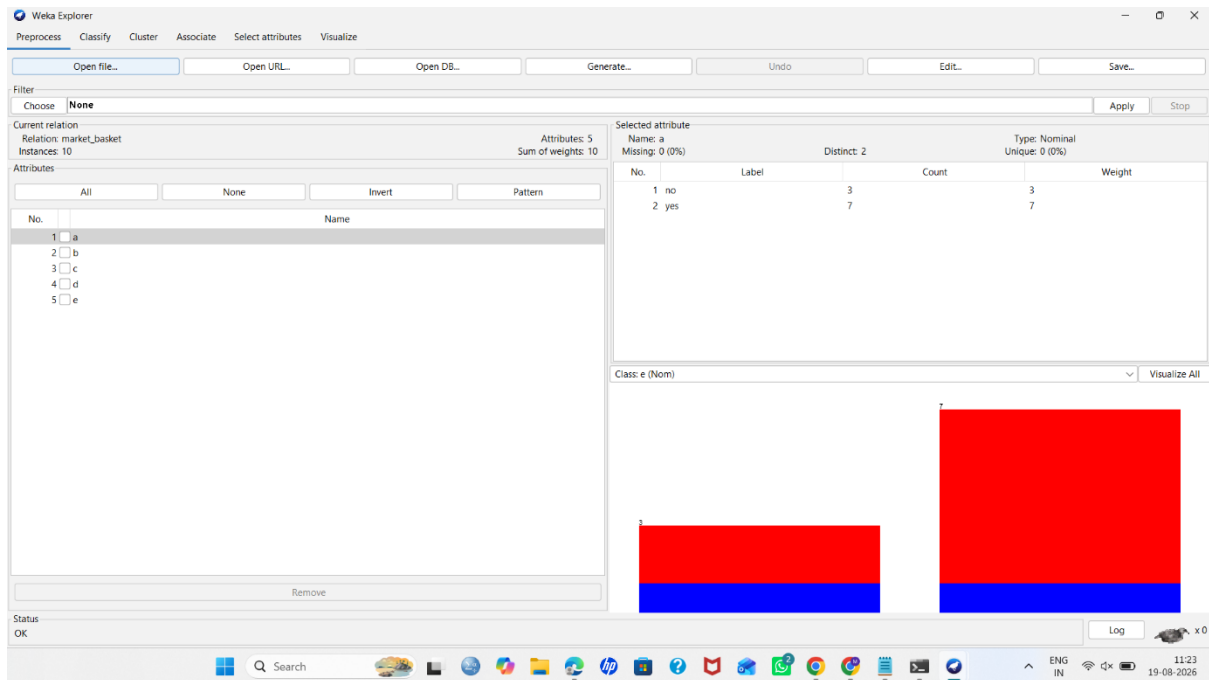


UNIT-3 LAB EXPERIMENTS

1.



2.

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Open file... Open URL... Open DB... Generate... Undo Edit... Save...

Filter: Choose **None** Apply Stop

Current relation: market_basket
Instances: 10
Attributes: 6 Sum of weights: 10

Attributes: All None Invert Pattern

No.	Name
1	☒ Milk
2	☐ Beer
3	☐ Diapers
4	☐ Bread
5	☐ Butter
6	☐ Cookies

Remove

Status: OK

Selected attribute: Name: Milk
Missing: 0 (0%) Distinct: 2 Type: Nominal Unique: 0 (0%)

No.	Label	Count	Weight
1	yes	5	5
2	no	5	5

Class: Cookies (Nom) Visualize All

Log 11:45 19-08-2026

Weka Explorer

Preprocess Classify Cluster **Associate** Select attributes Visualize

Associator: Choose **Apriori** -N 20 -T 0 -C 0.5 -D 0.05 -U 1.0 -M 0.3 -S -1.0 -c -1

Start Stop

Result list (right-click for...): 11:47:36 - Apriori

Associator output

```

=== Run information ===

Scheme: weka.associations.Apriori -N 20 -T 0 -C 0.5 -D 0.05 -U 1.0 -M 0.3 -S -1.0 -c -1
Relation: market_basket
Instances: 10
Attributes: 6
Milk
Beer
Diapers
Bread
Butter
Cookies

=== Associator model (full training set) ===

Apriori
=====

Minimum support: 0.45 (4 instances)
Minimum metric <confidence>: 0.5
Number of cycles performed: 11

Generated sets of large itemsets:

Size of set of large itemsets L(1): 11
Size of set of large itemsets L(2): 16
Size of set of large itemsets L(3): 6
Size of set of large itemsets L(4): 1

Best rules found:

1. Bread=yes ==> Beer=no 5 <conf: (1)> lift: (1.67) lev: (0.2) [2] conv: (2)
2. Butter=yes 5 ==> Beer=no 5 <conf: (1)> lift: (1.67) lev: (0.2) [2] conv: (2)
3. Butter=yes 5 ==> Bread=yes 5 <conf: (1)> lift: (2) lev: (0.25) [2] conv: (2.5)
  
```

Status: OK

Log 11:47 19-08-2026

Weka Explorer

Preprocess **Classify** Cluster **Associate** Select attributes Visualize

Associator: Choose **Apriori** - N 20 - T 0 - C 0.5 - D 0.05 - U 1.0 - M 0.3 - S -1.0 - c -1

Start Stop

Result list (right-click for ...): 11:47:36 - Apriori

Associator output:

```

Apriori
*****

Minimum support: 0.45 (4 instances)
Minimum metric <confidence>: 0.5
Number of cycles performed: 11

Generated sets of large itemsets:

Size of set of large itemsets L(1): 11
Size of set of large itemsets L(2): 16
Size of set of large itemsets L(3): 6
Size of set of large itemsets L(4): 1

Best rules found:

1. Bread=yes 5 ==> Beer=no 5 <conf: (1)> lift: (1.67) lev: (0.2) [2] conv: (2)
2. Butter=yes 5 ==> Beer=no 5 <conf: (1)> lift: (1.67) lev: (0.2) [2] conv: (2)
3. Butter=yes 5 ==> Bread=yes 5 <conf: (1)> lift: (2) lev: (0.25) [2] conv: (2.5)
4. Bread=yes 5 ==> Butter=yes 5 <conf: (1)> lift: (2) lev: (0.25) [2] conv: (2.5)
5. Butter=no 5 ==> Bread=no 5 <conf: (1)> lift: (2) lev: (0.25) [2] conv: (2.5)
6. Bread=no 5 ==> Butter=no 5 <conf: (1)> lift: (2) lev: (0.25) [2] conv: (2.5)
7. Bread=yes Butter=yes 5 ==> Beer=no 5 <conf: (1)> lift: (1.67) lev: (0.2) [2] conv: (2)
8. Beer=no Butter=yes 5 ==> Bread=yes 5 <conf: (1)> lift: (2) lev: (0.25) [2] conv: (2.5)
9. Beer=no Bread=yes 5 ==> Butter=yes 5 <conf: (1)> lift: (2) lev: (0.25) [2] conv: (2.5)
10. Butter=yes 5 ==> Beer=no Bread=yes 5 <conf: (1)> lift: (2) lev: (0.25) [2] conv: (2.5)
11. Bread=yes 5 ==> Beer=no Butter=yes 5 <conf: (1)> lift: (2) lev: (0.25) [2] conv: (2.5)
12. Beer=yes 4 ==> Bread=no 4 <conf: (1)> lift: (2) lev: (0.2) [2] conv: (2)
13. Beer=yes 4 ==> Butter=no 4 <conf: (1)> lift: (2) lev: (0.2) [2] conv: (2)
14. Beer=yes Butter=no 4 ==> Bread=no 4 <conf: (1)> lift: (2) lev: (0.2) [2] conv: (2)
15. Beer=yes Bread=no 4 ==> Butter=no 4 <conf: (1)> lift: (2) lev: (0.2) [2] conv: (2)
16. Beer=yes 4 ==> Bread=no Butter=no 4 <conf: (1)> lift: (2) lev: (0.2) [2] conv: (2)
17. Bread=yes Cookies=no 4 ==> Beer=no 4 <conf: (1)> lift: (1.67) lev: (0.16) [1] conv: (1.6)

```

Status: OK

Log

3.

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Open file... Open URL... Open DB... Generate... Undo Edit... Save...

Filter: Choose **None** Apply Stop

Current relation: Relation: buys_computer Instances: 14

Attributes: All None Invert Pattern

Attributes:

No.	Name
1	age
2	income
3	student
4	credit_rating
5	buys_computer

Remove

Selected attribute: Name: age Missing: 0 (0%) Distinct: 3 Type: Nominal Unique: 0 (0%)

No.	Label	Count	Weight
1	<=30	5	5
2	31..40	4	4
3	>40	5	5

Class: buys_computer (Nom) Visualize All

Status: OK

Log

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Classifier: Choose **ZeroR**

Test options:
☐ Use training set
☐ Supplied test set
☒ Cross-validation Folds: **10**
☐ Percentage split % **66**
 More options...

(Nom) buys_computer
 Start Stop

Result list (right-click for options):
 11:55:11 - rules.ZeroR

Classifier output:

```

income
student
credit_rating
buys_computer
Test mode: 10-fold cross-validation

=== Classifier model (full training set) ===

ZeroR predicts class value: yes

Time taken to build model: 0 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances      9      64.2857 %
Incorrectly Classified Instances    5      35.7143 %
Kappa statistic                    0
Mean absolute error                0.4962
Root mean squared error            0.4934
Relative absolute error            100 %
Root relative squared error        100 %
Total Number of Instances         14

=== Detailed Accuracy By Class ===
               TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Area  Class
               -----  -----  -
1.000   1.000   0.643    1.000   0.783    ?    0.178   0.555   yes
0.000   0.000    ?    0.000    ?    ?    0.178   0.318   no
Weighted Avg.   0.643   0.643    ?    0.643    ?    ?    0.178   0.470

=== Confusion Matrix ===
  a b  <-- classified as
  9 0 | a = yes
  5 0 | b = no
  
```

Status: OK

4.

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Open file... Open URL... Open DB... Generate... Undo Edit... Save...

Filter: Choose **None** Apply Stop

Current relation:
 Relation: diabetes
 Instances: 16

Attributes: 6 Sum of weights: 16

Attributes: All None Invert Pattern

No.	Name
1	Age
2	Glucose
3	BloodPressure
4	BMI
5	Insulin
6	Diabetes

Remove

Selected attribute:
 Name: Age
 Missing: 0 (0%)
 Distinct: 16
 Type: Numeric
 Unique: 16 (100%)

Statistic	Value
Minimum	21
Maximum	68
Mean	45.188
StdDev	14.73

Class: Diabetes (Num) Visualize All

Status: OK

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: Choose **LinearRegression** -S 0 -R 1.0E-8 -num-decimal-places 4

Test options:
☐ Use training set
☐ Supplied test set
☒ Cross-validation Folds: **10**
☐ Percentage split % **66**
 More options...

(Num) Diabetes ▾
 Start Stop

Result list (right-click for options):
14:19:00 - functions.LinearRegression

Classifier output:

```

=== Run information ===

Scheme:      weka.classifiers.functions.LinearRegression -S 0 -R 1.0E-8 -num-decimal-places 4
Relation:    diabetes
Instances:   16
Attributes:  6
  Age
  Glucose
  BloodPressure
  BMI
  Insulin
  Diabetes
Test mode:   10-fold cross-validation

=== Classifier model (full training set) ===

Linear Regression Model

Diabetes =
    0.0341 * BloodPressure +
    -2.3458

Time taken to build model: 0.05 seconds

=== Cross-validation ===
=== Summary ===

Correlation coefficient      0.6808
Mean absolute error         0.2806
Root mean squared error     0.3183
Relative absolute error     71.0518 %
Root relative squared error 69.7455 %
Total Number of Instances   16
  
```

Status: OK Log

36°C Partly sunny 21-08-2026 14:19

5.

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Open file... Open URL... Open DB... Generate... Undo Edit... Save...

Filter: Choose **None** Apply Stop

Current relation:
 Relation: transactions
 Instances: 5

Attributes: 11 Sum of weights: 5

Attributes: All None Invert Pattern

No.	Name
1	M
2	O
3	N
4	K
5	E
6	Y
7	D
8	A
9	U
10	C
11	I

Remove

Selected attribute:
 Name: M
 Missing: 0 (0%)
 Distinct: 2
 Type: Nominal
 Unique: 0 (0%)

No.	Label	Count	Weight
1	0	2	2
2	1	3	3

Class: I (Nom) Visualize All

Status: OK Log

36°C Partly sunny 21-08-2026 14:20

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Associate

Choose Apriori -I -N 10 -T 0 -C 0.8 -D 0.05 -U 0.5 -M 0.5 -S -1.0 -c -1

Start Stop

Result list (right-click for ...)

142433 - Apriori

Associate output

```
Run information ===
Scheme: weka.associations.Apriori -I -N 10 -T 0 -C 0.8 -D 0.05 -U 0.5 -M 0.5 -S -1.0 -c -1
Relation: transactions
Instances: 5
Attributes: 11
M
O
N
E
E
Y
D
A
U
C
I

=== Associate model (full training set) ===

Apriori
=====

Minimum support: 0.7 (3 instances)
Minimum metric <confidence>: 0.8
Number of cycles performed: 6

Generated sets of large itemsets:

Size of set of large itemsets L(1): 11

Large Itemsets L(1):
M=1 3
O=1 3
N=0 3
E=1 5
E=1 4
...
```

Status OK

Log

14:24 21-08-2026

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Associate

Choose Apriori -I -N 10 -T 0 -C 0.8 -D 0.05 -U 0.5 -M 0.5 -S -1.0 -c -1

Start Stop

Result list (right-click for ...)

142433 - Apriori

Associate output

```
M=1 3
O=1 3
N=0 3
E=1 5
E=1 4
Y=1 3
D=0 4
A=0 4
U=0 4
O=0 3
I=0 4

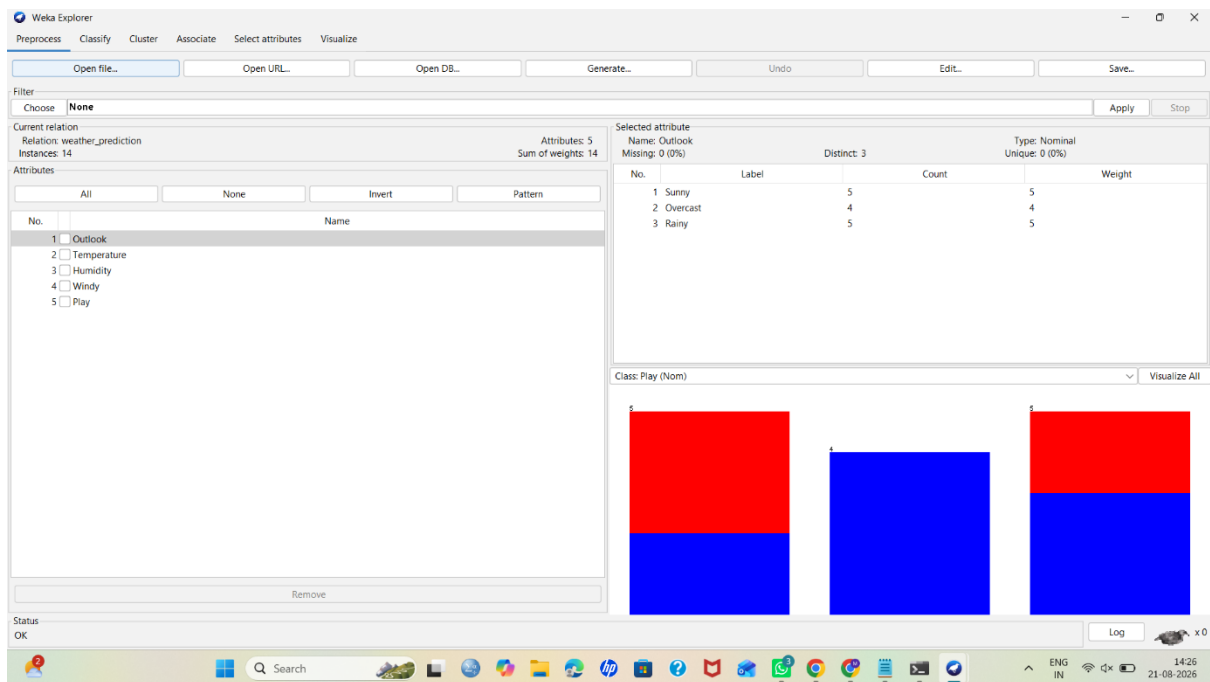
Size of set of large itemsets L(2): 31

Large Itemsets L(2):
M=1 E=1 3
M=1 D=0 3
M=1 I=0 3
O=1 E=1 3
O=1 E=1 3
O=1 A=0 3
O=1 D=0 3
N=0 E=1 3
N=0 D=0 3
E=1 E=1 4
E=1 Y=1 3
E=1 D=0 4
E=1 A=0 4
E=1 U=0 4
E=1 O=0 3
E=1 I=0 4
E=1 D=0 3
E=1 A=0 3
E=1 D=0 4
E=1 O=0 3
E=1 I=0 3
Y=1 A=0 3
Y=1 I=0 3
```

Status OK

Log

14:24 21-08-2026



Weka Explorer
Preprocess Classify Cluster Associate Select attributes Visualize

Classifier: Choose **J48 -C 0.25 -M 2**

Test options
☐ Use training set
☐ Supplied test set
☒ Cross-validation Folds: **10**
☐ Percentage split % **66**
More options...

(Nom) Play
Start Stop

Result list (right-click for options)
1427:17 - trees.J48

Classifier output

```
Humidity = Normal: Yes (2.0)
Outlook = Overcast: Yes (4.0)
Outlook = Rainy
Windy = True: No (2.0)
Windy = False: Yes (3.0)

Number of Leaves : 5
Size of the tree : 8

Time taken to build model: 0 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances 7 50 %
Incorrectly Classified Instances 7 50 %
Kappa statistic -0.0426
Mean absolute error 0.4167
Root mean squared error 0.5984
Relative absolute error 87.5 %
Root relative squared error 121.2987 %
Total Number of Instances 14

=== Detailed Accuracy By Class ===

      TP Rate  PP Rate  Precision  Recall  F-Measure  MDC  ROC Area  FRC Area  Class
0.556  0.600  0.625  0.556  0.588  -0.043  0.633  0.758  Yes
0.400  0.444  0.333  0.400  0.364  -0.043  0.633  0.457  No
Weighted Avg. 0.500  0.544  0.521  0.500  0.508  -0.043  0.633  0.650

=== Confusion Matrix ===

a b  <-- classified as
5 4 | a = Yes
3 2 | b = No
```

Status
OK

Log

Weka Explorer
Preprocess Classify Cluster Associate Select attributes Visualize

Classifier: Choose **Logistic -R 1.0E-8 -M -1 -num-decimal-places 4**

Test options
☐ Use training set
☐ Supplied test set
☒ Cross-validation Folds: **10**
☐ Percentage split % **66**
More options...

(Nom) Play
Start Stop

Result list (right-click for options)
1427:17 - functions.Logistic

Classifier output

```
=== Run information ===

Scheme: weka.classifiers.functions.Logistic -R 1.0E-8 -M -1 -num-decimal-places 4
Relation: weather_prediction
Instances: 14
Attributes: 5
  Outlook
  Temperature
  Humidity
  Windy
  Play
Test mode: 10-fold cross-validation

=== Classifier model (full training set) ===

Logistic Regression with ridge parameter of 1.0E-8
Coefficients...

Variable      Class
=====
Outlook=Sunny -45.2378
Outlook=Overcast 57.5375
Outlook=Rainy -5.9067
Temperature=Hot -8.3327
Temperature=Mild 44.8546
Temperature=Cool -45.4529
Humidity=Normal 118.1425
Windy=False 72.9648
Intercept -89.2032

Odds Ratios...

Variable      Class
=====
Outlook=Sunny 0
Outlook=Overcast 5.73275593611619E24
```

Status
OK

Log

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Classifier: Choose **Logistic** R 1.0E-8 M -1 -num-decimal-places 4

Test options:
☐ Use training set
☐ Supplied test set
☒ Cross-validation Folds: **10**
☐ Percentage split %: **66**
 More options...

(Nom) Play
 Start Stop

Result list (right-click for options):
 1427:17 - trees.J48
1427:41 - functions.Logistic

Classifier output:
 Outlook=Rainy 9.73275593611619E24
 Outlook=Rainy 0.0027
 Temperature=Hot 0.0002
 Temperature=Wild 3.0207875213740724E19
 Temperature=Cool 0
 Humidity=Normal 2.0353933107400553E51
 Windy=False 4.877521304260806E31

Time taken to build model: 0.02 seconds

=== Stratified cross-validation ===
 === Summary ===
 Correctly Classified Instances 10 71.4286 %
 Incorrectly Classified Instances 4 28.5714 %
 Kappa statistic 0.3778
 Mean absolute error 0.2858
 Root mean squared error 0.5345
 Relative absolute error 60.009 %
 Root relative squared error 108.3425 %
 Total Number of Instances 14

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
Weighted Avg.	0.778	0.400	0.778	0.778	0.778	0.378	0.767	0.817	Yes
	0.600	0.222	0.600	0.600	0.600	0.378	0.767	0.655	No

=== Confusion Matrix ===
 a b <-- classified as
 7 2 | a = Yes
 2 3 | b = No

Status: OK

7.

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Open file... Open URL... Open DB... Generate... Undo Edit... Save...

Filter: Choose **None** Apply Stop

Current relation: shopping_transactions
 Instances: 6

Attributes: Attributes: 5 Sum of weights: 6

All None Invert Pattern

No. Name

1 ☒ Hot_Dogs
 2 ☐ Buns
 3 ☐ Ketchup
 4 ☐ Coke
 5 ☐ Chips

Remove

Selected attribute: Name: Hot_Dogs Missing: 0 (0%) Distinct: 2 Type: Nominal Unique: 0 (0%)

No.	Label	Count	Weight
1	T	4	4
2	F	2	2

Class: Chips (Nom) Visualize All

Status: OK

Weka Explorer
Preprocess Classify Cluster Associate Select attributes Visualize

Associate
Choose **Apriori** -N 20 -T 0 -C 0.6 -D 0.05 -U 0.3333 -M 0.3333 -S -1.0 -c -1

Start Stop
Result list (right-click for ...)
14:31:15 - Apriori

Associate output

```

=== Run information ===

Scheme:      weka.associations.Apriori -T -N 20 -T 0 -C 0.6 -D 0.05 -U 0.3333 -M 0.3333 -S -1.0 -c -1
Relation:    shopping_transactions
Instances:    6
Attributes:   5
    Hot_Dogs
    Buns
    Ketchup
    Coke
    Chips

=== Associator model (full training set) ===

Apriori
=====

Minimum support: 0.4 (2 instances)
Minimum metric (confidence): 0.6
Number of cycles performed: 12

Generated sets of large itemsets:

Size of set of large itemsets L(1): 10

Large Itemsets L(1):
Hot_Dogs=T 4
Hot_Dogs=F 2
Buns=T 2
Buns=F 4
Ketchup=T 2
Ketchup=F 4
Coke=T 3
Coke=F 3
Chips=T 4
Chips=F 2

```

Status
OK

Log

Weka Explorer
Preprocess Classify Cluster Associate Select attributes Visualize

Associate
Choose **Apriori** -N 20 -T 0 -C 0.6 -D 0.05 -U 0.3333 -M 0.3333 -S -1.0 -c -1

Start Stop
Result list (right-click for ...)
14:31:15 - Apriori

Associate output

```

Large Itemsets L(4):
Hot_Dogs=T Buns=T Coke=F Chips=F 2
Hot_Dogs=T Buns=F Ketchup=F Coke=T 2
Hot_Dogs=T Buns=F Ketchup=F Chips=T 2
Hot_Dogs=T Buns=F Coke=T Chips=T 2
Hot_Dogs=T Ketchup=F Coke=T Chips=T 2
Buns=F Ketchup=F Coke=T Chips=T 3

Size of set of large itemsets L(5): 1

Large Itemsets L(5):
Hot_Dogs=T Buns=F Ketchup=F Coke=T Chips=T 2

Best rules found:

1. Buns=T 2 ==> Hot_Dogs=T 2 <conf: (1)> lift: (1.5) lev: (0.11) [0] conv: (0.67)
2. Chips=F 2 ==> Hot_Dogs=T 2 <conf: (1)> lift: (1.5) lev: (0.11) [0] conv: (0.67)
3. Hot_Dogs=F 2 ==> Buns=F 2 <conf: (1)> lift: (1.5) lev: (0.11) [0] conv: (0.67)
4. Hot_Dogs=F 2 ==> Chips=T 2 <conf: (1)> lift: (1.5) lev: (0.11) [0] conv: (0.67)
5. Buns=T 2 ==> Coke=F 2 <conf: (1)> lift: (2) lev: (0.17) [1] conv: (1)
6. Chips=F 2 ==> Buns=T 2 <conf: (1)> lift: (3) lev: (0.22) [1] conv: (1.33)
7. Buns=T 2 ==> Chips=F 2 <conf: (1)> lift: (3) lev: (0.22) [1] conv: (1.33)
8. Ketchup=T 2 ==> Coke=F 2 <conf: (1)> lift: (2) lev: (0.17) [1] conv: (1)
9. Chips=F 2 ==> Coke=T 2 <conf: (1)> lift: (2) lev: (0.17) [1] conv: (1)
10. Buns=T Coke=F 2 ==> Hot_Dogs=T 2 <conf: (1)> lift: (1.5) lev: (0.11) [0] conv: (0.67)
11. Hot_Dogs=T Coke=F 2 ==> Buns=T 2 <conf: (1)> lift: (3) lev: (0.22) [1] conv: (1.33)
12. Hot_Dogs=T Buns=T 2 ==> Coke=F 2 <conf: (1)> lift: (2) lev: (0.17) [1] conv: (1)
13. Buns=T 2 ==> Hot_Dogs=F Coke=F 2 <conf: (1)> lift: (3) lev: (0.22) [1] conv: (1.33)
14. Buns=T Chips=F 2 ==> Hot_Dogs=T 2 <conf: (1)> lift: (1.5) lev: (0.11) [0] conv: (0.67)
15. Hot_Dogs=T Chips=F 2 ==> Buns=T 2 <conf: (1)> lift: (3) lev: (0.22) [1] conv: (1.33)
16. Hot_Dogs=T Buns=T 2 ==> Chips=F 2 <conf: (1)> lift: (3) lev: (0.22) [1] conv: (1.33)
17. Chips=F 2 ==> Hot_Dogs=T Buns=T 2 <conf: (1)> lift: (3) lev: (0.22) [1] conv: (1.33)
18. Buns=T 2 ==> Hot_Dogs=T Chips=F 2 <conf: (1)> lift: (3) lev: (0.22) [1] conv: (1.33)
19. Hot_Dogs=T Buns=F 2 ==> Ketchup=F 2 <conf: (1)> lift: (1.5) lev: (0.11) [0] conv: (0.67)
20. Hot_Dogs=T Coke=T 2 ==> Buns=F 2 <conf: (1)> lift: (1.5) lev: (0.11) [0] conv: (0.67)

```

Status
OK

Log